

Problem Solving

1. Two independent samples of sizes $n_1 = 6$ and $n_2 = 10$ are taken from two normal population with equal variances. Find b such that $P(\frac{s_1^2}{s_2^2} < b) = 0.95$.

2. Let $X_1, \dots, X_n$ be a random sample from $N(\mu, \sigma^2)$, where both parameters μ and σ^2 are unknown. How to construct a confidence interval for σ^2 ? If μ is known, how would you modify the procedure for finding a confidence interval for σ^2 ?

3. Let $X_1, \dots, X_{n_1}$ be a random sample from $N(\mu_1, \sigma_1^2)$ and $Y_1, \dots, Y_{n_2}$ be a random sample from $N(\mu_2, \sigma_2^2)$.

- (a) Construct a confidence interval for $\mu_1 - \mu_2$ when σ_1^2 and σ_2^2 are known.
- (b) Construct a confidence interval for $\mu_1 - \mu_2$ when $\sigma_1^2 = \sigma_2^2 = \sigma^2$.

